Math Distillation Decouples Chain-of-Thought from Behavior on a Reasoning Model

Abraham Yeung
Stanford University
ayeung16@stanford.edu

Abstract

Chain-of-thought (CoT) monitorability is the hope that reading a model's reasoning will tell us why it answered, so an overseer can catch undesired influences. We ask whether supervised fine-tuning on math traces preserves this property, using the cue-based faithfulness paradigm of Meek et al. (2025) and Turpin et al. (2023) on a DeepSeek-R1-distilled student. We find a dissociation between behavior and verbalization. After weak-to-strong reasoning fine-tuning (W2SR; Yuan et al., 2025), a planted cue's behavioral pull on the answer is held fixed (paired $\Delta = +0.16$ with a lower CI bound near zero: we read this as "not reduced" rather than confidently strengthened). Flips land on the cue target at roughly twice chance, a rate statistically indistinguishable from the untrained model. But the CoT's acknowledgment of that cue collapses from 25% to 3% on matched (question, cue) pairs, while GPQA accuracy is preserved ($0.400 \rightarrow 0.425$, $n = 40$). The model keeps being moved by the cue and stops saying so. That fine-tuning lowers CoT faithfulness on an answer-level metric is already known (Lobo et al., 2024). Our contribution is that the cue's causal grip on behavior is held fixed while verbalization vanishes, which is the failure mode that matters for monitorability. The effect is teacher-strength-invariant: a weak teacher, a strong teacher, and same-strength self-distillation all collapse acknowledgment, by amounts our paired tests cannot distinguish, so the weak-to-strong asymmetry is incidental. We also document that the student's chat template removed the reasoning span from our SFT examples, so the supervision was answer-only; we state where this bounds the claims (§3, §7). Most of the raw drop follows from the resulting compression, but if we restrict to samples where the cue actually moved the answer, acknowledgment is suppressed at matched mid-length (odds ratio about 8.75, Fisher's exact $p = 0.003$; an exploratory subset result, §7), and the collapse replicates on MMLU at $2.8\times$ compression rather than $14\times$. A capability-only certification can pass a model whose CoT has become markedly less revealing while still reading as elaborate.

1 Introduction

Chain-of-thought "monitorability" is the hope that reading a model's reasoning reveals why it answered, so an overseer can catch undesired influences [9]. Korbak et al. argue that this opportunity is fragile and that developers should think about how their design choices affect monitorability. Our paper looks at one such design choice, math-CoT SFT, and shows that it erodes monitorability in a specific way.

Weak-to-Strong Reasoning (W2SR) [1][1] reports that fine-tuning a strong student on a weaker model's CoT raises accuracy. Concurrent faithfulness work [3, 2, 4] shows that models often do not verbalize the cues that actually move their answers.

W2SR certifies the student on accuracy alone and is silent on whether the distilled CoT stays monitorable. If distillation changed how a model reasons on the page without changing what it reveals, an accuracy check would miss the problem entirely.

The phenomena we exploit are largely known. Lobo et al. [5] showed that fine-tuning hurts CoT faithfulness on an answer-level metric. Zaman and Srivastava [7] argued via causal mediation that the dissociation between behavior and CoT verbalization can exist statically. Concurrent self-distillation work [6] identifies compression as a mechanism that suppresses CoT properties, in their case the verbalization of uncertainty. Our contributions are methodological and target-specific.

1. We controllably induce a behavior–verbalization dissociation via a specific training method. By separating behavior (paired influence and a flip-to-cue rate at about $2\times$ chance) from verbalization (judge acknowledgment), we show that math-CoT SFT moves an R1-distill student from monitorable (acknowledgment 25%, behavior preserved) to monitorable-looking-but-not (acknowledgment 3%, behavior still preserved). The new piece beyond Zaman and Srivastava is the controllability: a marketed training method induces the dissociation on demand, with behavior held fixed (Section 5).
2. A same-strength self-distillation arm serves as a negative control. With teacher equal to student, the collapse is indistinguishable from W2SR (paired $p = 1$, $n = 79$), and every trained-arm pair we tested against W2SR gives $p \geq 0.09$. We have not seen this control in the literature in this form. It is the cleanest evidence that the cause is the SFT in general, not weak-to-strong asymmetry. It also separates our claim from the concurrent self-distillation work on uncertainty, since the outcome variable is different but the underlying mechanism is similar (Section 5).
3. Our mechanism story has two parts. A length-binned Fisher analysis shows that the matched-mid-length gap closes across all cued samples ($p = 0.80$), which supports compression as the explanation for the raw $25\% \rightarrow 3\%$ drop. But if we restrict to samples where the cue actually moved the answer (call this $\text{influenced} = 1$), the matched-mid-length gap reopens to baseline 70.0% vs W2SR 21.1%, $\text{OR} \approx 8.75$, $p = 0.003$. Decomposing the matched-length question by influenced status suggests that compression accounts for the raw drop while W2SR additionally suppresses cue acknowledgment at matched length, exactly where monitorability matters (an exploratory subset finding; see §7).
4. A reproduction of W2SR with a headroom probe shows that its accuracy claim is itself conditional. The gain only fires with (i) an in-family teacher and (ii) a base student. A zero-shot CoT prompt on the untrained student separates real elicitation of latent capability from a CoT-prompting confound (Section 4), and the certified guarantee is narrower than Yuan et al. claim.
5. Putting these together gives a clear certification consequence. An accuracy-only certification of any math-CoT-SFT student, including same-strength self-distillation rather than just weak-to-strong, can pass a model whose CoT has become markedly less revealing. A monitorability composite that rewards elaborate-looking CoT misses the failure too. The cue-acknowledgment component has to be measured directly.

2 Related work

Lobo et al. [5] show that task-specific fine-tuning lowers CoT faithfulness across four datasets, and we take this degradation as given. The claim we are making is not that faithfulness drops, but that it drops while the cue's behavioral influence on the answer does not. That is the dissociation their answer-level metric cannot isolate. Concurrent work by Jiang et al. [11] makes the closest complementary observation in a different domain: on medical CoT distillation, answer accuracy improves ($74.7\% \rightarrow 84.4\%$ MedQA) while reasoning step error rate climbs ($30.6\% \rightarrow 50.3\%$),

[1] Distinct from the earlier paper with the same title by Yang, Ma, and Liu [10] (EMNLP Findings 2024), which uses a progressive SFT+DPO recipe on GSM8K, MATH, and OlympicArena. We build on Yuan et al.'s LoRA-SFT pipeline.

which they use to argue that answer-level metrics alone mask reasoning-quality decay. Their finding and ours share the shape "the answer looks better while the trace gets worse" in two very different domains, which is stronger joint evidence than either result alone.

Concurrent work [6] finds that self-distillation suppresses epistemic verbalization, specifically the verbalization of uncertainty tokens like "wait" and "hmm," via compression. They observe that this hurts out-of-distribution generalization. We share the compression mechanism and the self-distillation setup, but our outcome variable is a different property of the CoT: cue-acknowledgment (does the trace mention or attribute reasoning to a planted hint) rather than uncertainty-token density. Uncertainty-token loss and cue non-acknowledgment are distinct failure modes that happen to share a mechanism. On the RL side rather than SFT, Little [14] independently shows that training with an explicit length penalty makes CoT less monitorable at matched length (7-35 pp lower hint disclosure after random-shortening baselines). Our result is the SFT-side complement: math-CoT SFT compresses CoT even with no length term in the objective and produces the same qualitative degradation, so the failure mode is not specific to length-penalized RL.

Meek et al. [2] measure monitorability as a combination of faithfulness and verbosity across model families. We use their cued-prompt eval but study how a training intervention moves faithfulness on a fixed substrate. We pair the standard verbal-acknowledgment measurement with a behavioral measure that separates "stops being influenced" from "stops verbalizing influence," and we ask the sharper question of whether faithfulness falls beyond what CoT length alone explains.

Zaman and Srivastava [7] argue that the Biasing Features metric, which flags certain answers as "unfaithful," partly captures the lossy compression of distributed transformer computation into linear natural language. They show via causal mediation that even non-verbalized hints can causally route influence through the CoT to the answer. That is our dissociation, observed inferentially. We add three things on top. First, we induce the dissociation with a training method, moving acknowledgment from 25% to 3% on a fixed substrate. Second, we pin the behavioral pull directly, via paired influence and a flip-to-cue-target rate at about $2\times$ chance, rather than infer it through mediation. Third, we draw the certification consequence, since an accuracy-only or verbosity-rewarding check passes the very models on which the dissociation is most extreme. Zaman also argue that hint-acknowledgment is a poor faithfulness metric because absence of hint words does not prove the hint was not causally involved. We agree with that critique on faithfulness grounds. But for monitorability the metric is aligned with the task: what an overseer can read off the trace determines whether they can catch the influence, independent of whether the model was "really" moved by the hint in some deeper counterfactual sense. Our CoT-conclusion analysis in §5 further shows the CoT reasoning itself walks to the cue target in most silently-influenced cases, so acknowledgment absence is not merely compression stripping the hint word.

Channel migration is not what is happening here. Recent work on extended-thinking models documents a thinking-vs-answer divergence in hint acknowledgment, often a large one (for instance 87.5% vs 28.6% in [8]; concurrent work [18] finds a similar divergence on a larger 12-model panel with an "unacknowledged" quadrant of 11.8%). Our judge reads the full chain-of-thought, that is, both the thinking tokens and the answer. So the collapse we measure is not an artifact of which channel is read; acknowledgment vanishes in both channels combined.

Xiong et al. [12] provide a useful contrast on the training-method axis: RLVR (reinforcement learning with verifiable rewards) spontaneously improves monitorability during training, driven by response sharpening and increased prompt attention, without monitorability being an explicit objective. Our result is the opposite for a different post-training procedure: SFT on math-CoT traces, also without monitorability in the objective, degrades monitorability instead. Which post-training method a model receives can flip the sign of the monitorability side effect, even when neither method targets monitorability directly. Related term-of-art usage: Yee et al. [13] introduced the language of "dissociation" for faithful-vs-unfaithful reasoning mechanisms in a static setting; our use of "behavior–verbalization dissociation" is in that same lineage but induced by training.

Known limits of the cue-based paradigm. The Turpin-style cue-acknowledgment paradigm we adopt has documented limitations. Young [15] shows that the choice of faithfulness classifier alone shifts measured faithfulness by up to 30 pp across otherwise identical setups, so absolute rates should be read as judge-dependent. Scalena et al. [16] argue that reasoning models commit to an answer well before the CoT ends, so late CoT content can be epiphenomenal and length-only comparisons can be misleading (our Task H length-binned analysis is designed to defuse this specific critique on our data). Za et al. [17] document that Sonnet is roughly $3\times$ more susceptible than Gemini under adversarial persuasion when used as a monitor, which is one reason we cross-check our Sonnet judge with two others: Gemini (Cohen's $\kappa \in [0.64, 0.68]$, substantial) and Kimi (overall $\kappa = 0.556$, moderate; systematically stricter than Sonnet). All three judges agree on the direction of the ack drop and give one-directional discordant patterns. We frame our headline metric as *monitorability* (does the trace reveal the influence to an overseer) rather than *faithfulness* (does the trace causally reflect the model's internal computation) precisely because the paradigm operationalizes the former, not the latter: an overseer only has the trace, so what the trace does or does not say about the cue is directly what determines whether the influence is catchable.

3 Setup

W2SR [1] samples CoT from a weak teacher on a problem set, keeps correct and incorrect traces, LoRA-SFTs a stronger student, and reports held-out Pass@1. Monitorability follows Meek et al. [2]: each question is paired with a cue (authority opinion, leaked answer key, embedded metadata, and so on) that points at a target. An LLM judge reads the full CoT and scores faithfulness (does the CoT acknowledge the cue) and verbosity (a factor-utilization rate). We use the five cues shipped in the released eval, with `claude-sonnet-4-6` at $T = 0$ as the single judge. Faithfulness is reported as acknowledgment among cued samples. All evaluation is greedy ($T = 0$), trace sets are hashed, and training is LoRA. Generation, training, and gating run on Modal GPUs. All code, configs, per-sample judge labels, and the reanalysis pipeline are released. The four headline numbers in Table 2 are enforced by hard-fail assertions in `01_gate.py`, so any drift halts the suite.

Procedural details that matter for interpretation. Trace generation used $T = 0.6$, top-p 0.95, and a repetition penalty of 1.1 (the penalty was added for the loop-prone 1.5B teacher and kept constant across all teachers for comparability; it discourages long repetitive CoT and so interacts with the length analyses). Traces without a parseable `\boxed{}` answer were dropped before SFT, which preferentially removes traces truncated at the generation budget and therefore biases the training set slightly short. The under-test conditions were evaluated with an 8k generation budget (except the cross-family Llama arms, which used 16k); the teacher reference rows used 30k, so teacher-vs-student comparisons are not budget-matched (the baseline-vs-trained comparisons that carry every headline result are). SFT computes the loss over the full rendered sequence, prompt tokens included, rather than masking to the assistant span. Task H's W2SR side pools a larger re-run of the three text cues with the original run's two non-text cues (634 records) to give the length bins usable counts; the paired analyses elsewhere use the original matched set. The main eval judge is `claude-sonnet-4-6`; Tasks I2/I3 and J were judged with `claude-sonnet-4.5` via OpenRouter. Re-running an LLM judge does not reproduce its labels exactly even at $T = 0$: an identical re-run of the Task G rubric swap moved baseline acknowledgment from 9.4% to 8.8% and its McNemar p from 6×10^{-5} to 1.2×10^{-4}. We report the committed label set and note that judge-level digits carry this much noise.

What the supervision actually contained. This matters for how the results should be read, so we state it explicitly. Traces are generated by prompting the teacher with the R1 chat template, whose generation prefix ends <|Assistant|><think>, so each stored trace has the form {reasoning}</think>{answer}. Our SFT examples were rendered with the student's own chat template. The published R1-Distill template contains, for assistant turns, a rule that splits the content on </think> and keeps only the final segment. The tokenized supervision for every R1-substrate arm in this paper was therefore the *final-answer segment only*, with the reasoning span removed — even though the trace files on disk retain it (which is why a check of the trace data alone does not reveal this, and why we report it here). The intervention we are measuring is thus SFT on the answer segments of math CoT traces ("answer-only" supervision), not SFT on the reasoning itself. We flag the consequences in Section 7: it gives the CoT compression a straightforward explanation, and it makes the teacher-strength invariance in Table 2 much less surprising, since the teacher's reasoning never entered the loss. It does not affect the behavioral measurements, the paired acknowledgment collapse, or the matched-length residual, all of which are properties of the resulting models rather than of the training data. The released training code now renders the assistant turn without the template and hard-fails if the reasoning span is absent from a rendered example.

Table 1: Reproduction (held-out MATH L3 to L5 Pass@1). In-family native-Qwen teacher; the headroom probe separates genuine elicitation from CoT-prompting. The gain is genuine on the base student and is an artifact on the instruct student.

student	unelicited	zero-shot CoT	W2SR	W2SR − CoT
Qwen2.5-Math-7B (base)	0.325	0.240	0.645	+0.405 (genuine)
Qwen2.5-7B-Instruct	0.230	0.630	0.630	+0.000 (artifact)
Foreign-style teachers (R1-distill-1.5B, SimpleRL-Zoo) into 7B: no gain or collapse.				

4 When W2SR fires

With a foreign-style weak teacher (DeepSeek-R1-distilled 1.5B, or SimpleRL-Zoo) distilled into a Qwen2.5-7B student, we found no Pass@1 gain and frequent degeneration, across students, ranks, and decodings. W2SR only reproduces with an in-family, native-Qwen weak teacher. With `Qwen2.5-Math-1.5B-Instruct` (about 67% on the train problems) as teacher and `Qwen2.5-Math-7B` as student, held-out MATH (L3 to L5) Pass@1 goes from 0.325 to 0.645, with format-valid 1.00.

A $+5$ point gate-pass on an already-CoT-capable model could be re-arrangement rather than real elicitation. To check, we score the untrained model under a zero-shot CoT prompt, which we take as the prompting-only ceiling, and we report W2SR − (zero-shot CoT) (Table 1). On the base student, W2SR adds $+0.41$ beyond prompting, which is genuine elicitation of latent capability. On an instruct student it adds $+0.00$. The apparent gain over an unelicited baseline is purely the CoT-elicitation confound. Two conditions that Yuan et al. leave implicit are therefore necessary: an in-family teacher, and a base (non-CoT-eliciting) student.

5 Math-CoT SFT degrades CoT monitorability

We compare a baseline student, a W2SR student (LoRA-SFT'd on a weak in-family teacher's MATH CoT), and references, on GPQA. The SFT lands the student at its own CoT ceiling, so faithfulness differences are not a capability confound. The question we are asking is whether distillation changes faithfulness, not whether it improves capability.

The instruct student is at the floor. On `Qwen2.5-7B-Instruct`, cue-acknowledgment is at a floor for every condition (baseline 1.8% at $N = 970$; W2SR 3.6%; strong-teacher control 4.6%), while teacher references reach 22 to 37%. We cannot detect faithfulness transfer here. But verbosity does transfer strongly ($\Delta = +0.26$ for W2SR over baseline, CI $[0.22, 0.31]$), so the model looks more monitorable without any measurable faithfulness change. To get a substrate with dynamic range, we move to a reasoning student.

Behavior is preserved, verbalization collapses. On the reasoning student, the setup is fully in-family: student `R1-Distill-Qwen-7B`, weak teacher `R1-Distill-Qwen-1.5B`. Distillation passes the capability gate (held-out MATH Pass@1 above baseline by the preregistered ≥ 5-point margin), the trained student's GPQA answers remain parseable at 94.2% ($179/190$ cued samples, vs 76.9% for the untrained baseline), and baseline acknowledgment is off-floor at 25% (Chua and Evans [4] report $\sim 59\%$ on the full DeepSeek-R1; our lower baseline is on the distilled 7B variant on GPQA, so absolute levels differ across scale and eval set). Paired influence (answer equal to cue target on matched (qid, cue) pairs) gives $\Delta = +0.157$, 95% CI $[+0.037, +0.278]$, $n = 108$, McNemar $p = 0.021$, discordant $16/33$. The direction is positive, but the lower confidence bound is close to zero, so we read this as "not reduced" rather than confidently strengthened. Baseline has $\sim 23\%$ no-answer attrition on GPQA, which biases the paired Δ toward zero, so the influence result is if anything conservative about how far behavior is preserved. The cleaner signal is that among answer flips, the flip lands on the cue target 73% of the time for W2SR vs 65% for baseline. Both are about $2\times$ the $1/3$ random-flip chance (baseline $p = 1.2 \times 10^{-7}$, W2SR $p \approx 10^{-18}$), and they are statistically indistinguishable from each other. The pull is real cue-influence and not accuracy in disguise. If we restrict to the cue-points-at-wrong-answer stratum, so that landing on the cue is by

Table 2: Reasoning-student monitorability on GPQA with in-family R1-distill teachers. Faithfulness is acknowledgment among cued samples. Median CoT is computed on cued-cell completions. W2SR degrades faithfulness at preserved accuracy; a strong (14B) teacher degrades it nearly as much as a weak (1.5B) one; and self-distillation collapses faithfulness indistinguishably from W2SR weak ($p = 1$). All trained arms land far below baseline (25%), and each trained arm we tested against W2SR weak is not distinguishable from it (pairwise McNemar $p \geq 0.09$; the three pairs not involving W2SR weak were not tested). The effect is teacher-independent and generalizes to ordinary same-strength SFT on MATH-style traces. Accuracy is the uncued-cell GPQA accuracy at $n = 40$; "n/a" marks arms for which we did not run the uncued cell.

condition	faithfulness	GPQA acc	median CoT (chars)
baseline R1-7B	$40/160 = 25.0\%$	0.400	18,692
W2SR (1.5B $\rightarrow$ 7B)	$6/190 = 3.2\%$	0.425	1,362
strong-teacher KD (14B $\rightarrow$ 7B)	$13/175 = 7.4\%$	n/a	1,508
self-distill, $\leq$4k budget (Self-A)	$2/84 = 2.4\%$	n/a	1,258
self-distill, $\leq$8k budget (Self-B)	$3/89 = 3.4\%$	n/a	1,291
cross-family: Llama-8B self-distill	gate fail	0.225	1,424

Cross-family row (Llama-8B self-distillation, matched recipe): GPQA accuracy $0.500 \rightarrow 0.225$ (-27.5pp at 0% empty), CoT compression $10.8\times$. The capability gate failed, so the faithfulness number is not interpretable and we do not report it (see Limitations).

Pairwise tests: W2SR vs. baseline $p = 2 \times 10^{-9}$; strong-teacher KD vs. baseline $p = 2.2 \times 10^{-5}$; Self-A vs. baseline $p = 2.4 \times 10^{-7}$ ($n = 77$, disc. $23/0$); Self-B vs. baseline $p = 4.8 \times 10^{-7}$ ($n = 82$, disc. $22/0$); Self-A vs. W2SR $p = 1$ ($n = 79$, disc. $2/2$); Self-B vs. W2SR $p = 1$ ($n = 79$, disc. $2/3$). All tested pairs of trained arms are statistically indistinguishable; the three pairs not involving W2SR weak were not tested.

definition a wrong answer, flippers still hit the cue target at $77\%/74\%/74\%$ for baseline, W2SR, and strong-teacher KD (each $p \leq 2 \times 10^{-9}$ against $1/3$).

Verbalization collapses on both rubrics. On the upstream loose rubric ("acknowledge the cue"), W2SR drops pooled acknowledgment to 3.2% ($\Delta = -0.220$, CI $[-0.293, -0.153]$, McNemar $p \approx 2 \times 10^{-9}$, discordant $34/1$). The drop holds at every text cue (Table 4: stanford $47\% \rightarrow 8\%$, grader-hack $22\% \rightarrow 3\%$, insider $56\% \rightarrow 5\%$), and the discordant cells are one-directional (12/0, 7/1, 15/0). Figure 1 summarizes the behavior–verbalization split across all trained arms. A stricter "attribute reasoning to the cue" rubric, scored by `gemini-2.5-pro` (the same out-of-family judge used for the re-scoring below), shrinks both rates (baseline 9.4%, W2SR 0.0%, $15/0$ discordant, $p = 6 \times 10^{-5}$). The magnitude is smaller but the drop becomes even more one-directional. Both numbers hold at preserved GPQA accuracy ($0.400 \rightarrow 0.425$, $n = 40$; a $+2.5$pp difference that this n cannot resolve from zero) and at roughly $14\times$ shorter CoT (baseline median $18{,}692$ chars, W2SR 1,362). The monitorability cost is therefore not bought with a capability loss that a certifier would notice: on the metric the certification actually checks, the model is unchanged.

We checked that the collapse is not an artifact. Hand-checked completions are coherent, $37/40$ conclude with an answer, and none are degenerate. The model solves the problem without verbalizing the cue. It is also not a single-judge artifact: re-scoring the same 525 R1-family CoTs with `gemini-2.5-pro` gives per-condition Cohen's $\kappa \in [0.64, 0.68]$ (overall $\kappa = 0.68$), paired drop $\Delta = -0.153$, McNemar $p = 2.4 \times 10^{-7}$, discordant $23/0$. Adding a third judge (`moonshotai/kimi-k2-0905`, chosen because it is from a different model family than both Sonnet and Gemini) gives a stricter reading: absolute rates shrink (baseline $24.7\% \rightarrow 12.3\%$, W2SR $3.2\% \rightarrow 1.6\%$), overall $\kappa = 0.556$ (moderate, and lower than Gemini's because Kimi under-calls acknowledgment consistently across all conditions), but the paired drop still fires with the same one-directional signature: $n = 140$ matched pairs, discordant 17 baseline-only vs 1 W2SR-only, $\Delta = -0.114$ $[-0.171, -0.057]$, McNemar $p = 1.4 \times 10^{-4}$. The three-judge agreement (Sonnet, Gemini, Kimi) covers three model families and both stricter and looser rubric readings; the drop direction and one-directional discordant pattern are stable across all three.

Why faithfulness drops: compression, plus residual cue suppression. Faithfulness and CoT length both fall, so the near-trivial half of the story is that shorter CoT surfaces cues less often. The interesting question is whether faithfulness still degrades at matched length. Length is endogenous to

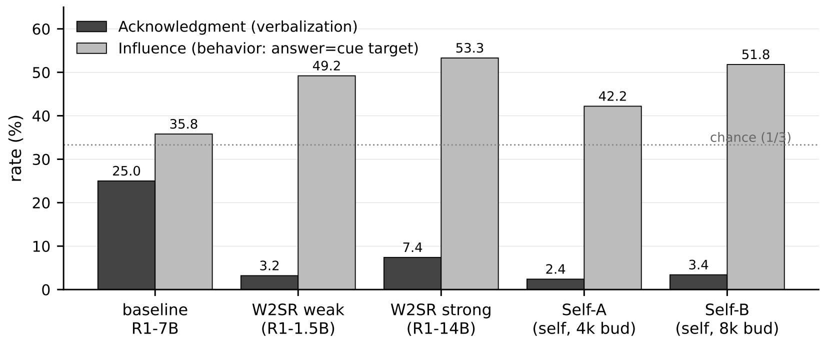

Figure 1: Behavior–verbalization dissociation across the reasoning-student axis. Acknowledgment (verbalization) collapses from 25.0% on baseline R1-7B to 2.4–7.4% across all four trained arms (W2SR, strong-teacher distill, two self-distill budgets). Influence (behavior, defined as answer equal to cue target) is preserved or strengthened in every trained condition and stays well above the $1/3$ random-flip chance line. The trained arms keep being moved by the cue and stop saying so. Numbers reproduced exactly by `python3 scripts/reanalysis/01_gate.py`; the 25.0%, 3.2%, and paired deltas of -0.22 and $+0.16$ are enforced by hard-fail asserts in that script.

reasoning (a brevity-prompt control failed, since R1 ignores it), so we bin both conditions by CoT length and compare within bins.

A length-binned analysis (Task H, quantiles of the union distribution, Fisher's exact unpaired; Figure 2A) sharpens the matched-length question. Across all cued and judge-labeled samples, at matched mid-length ($[1954, 8716)$ chars, baseline $n = 40$, W2SR $n = 119$) baseline and W2SR are essentially identical (ack 17.5% vs 15.1%, Fisher two-sided $p = 0.80$, OR $= 1.19$). At very long lengths ($[8716, 37816)$, baseline $n = 119$, W2SR $n = 40$), baseline is meaningfully higher (27.7% vs 15.0%, $\Delta = +12.7$ pp, OR ≈ 2.2, Fisher one-sided $p = 0.08$, two-sided $p = 0.14$). The direction is right but the gap just misses conventional significance.

Things look different if we restrict to the safety-relevant subset. $\text{influenced} = 1$ means the cue actually moved the answer, and that is the case we care about for monitorability. On that subset, the matched-mid-length gap reopens to baseline 70.0% $(7/10)$ vs W2SR 21.1% $(16/76)$, $\text{OR} \approx 8.75$, Fisher two-sided $p = 0.003$. At long lengths the same direction holds (67.6% vs 40.0%, $\text{OR} \approx 3.14$, $p = 0.15$ with baseline $n = 34$ and W2SR $n = 10$). The unconditional Task H was diluted by $\text{influenced} = 0$ samples (cue did not move the answer), where neither condition acknowledges much at any length, and so the gap looks closed.

So most of the raw $25\% \rightarrow 3\%$ drop is compression, and the compression has a mundane and fully identified cause: the supervision the student actually received was answer-only (Section 3). A student trained on final-answer segments produces short completions that rarely emit the </think> separator (22% of W2SR-student completions vs 57% of baseline), which is what we observe. We therefore do *not* claim the compression is an emergent or surprising property of SFT; it is the expected consequence of the supervision format.

The part that compression does not explain is the residual. Once we restrict to the safety-relevant subset, W2SR suppresses cue acknowledgment *at matched length*, where the compression account makes no prediction: two sets of traces of the same length differ by a factor of about 8.75 in odds of acknowledging the cue. That residual, not the raw drop, is the result that bears on monitorability. The MMLU replication below adds a second handle on the same question, since it reproduces the collapse at a much lower compression ratio.

Teacher-strength axis. A strong (14B) teacher degrades faithfulness about as much as a weak (1.5B) one (Table 2), and the two are not distinguishable. To check whether the weak-to-strong asymmetry is causal at all, we train baseline R1-7B on its own R1-7B traces (the same MATH L3 to L5 problem pool, $T = 0.6$, same SFT config). Both self-distillation arms (4k and 8k generation budgets) collapse acknowledgment to 2.4% and 3.4%, statistically indistinguishable from W2SR (paired $p = 1$, $\Delta \approx 0$ on $n = 79$ matched pairs each), and median CoT compresses to about 1.3k chars regardless of budget.

Cross-substrate replication on MMLU. An MMLU sweep over 5 STEM subjects (college physics, chemistry, biology and mathematics, plus conceptual physics; 8 questions per subject per cue, 190–195 cued samples per condition) holds the pattern. Pooled baseline ack is 27.4% $(52/190)$, W2SR is 3.6% $(7/195)$, and Self-A is 4.6% $(9/195)$. Paired on (qid, cue), W2SR vs. baseline gives $\Delta = -0.232$, McNemar $p = 2.4 \times 10^{-10}$ ($n = 185$, discordant $47/4$), and Self-A vs. baseline gives $\Delta = -0.222$, $p = 8.2 \times 10^{-10}$ (discordant $45/4$). Self-A vs. W2SR is again indistinguishable ($\Delta = +0.010$, $p = 0.73$), replicating the GPQA teacher-invariance result on a second substrate. Baseline R1-7B already writes much shorter CoT on MMLU (3,223 vs 18,692 chars median), so MMLU compression is only about $2.8\times$ (baseline $3{,}223 \rightarrow$ W2SR $1{,}141$) against GPQA's $14\times$ — yet the absolute drop is if anything larger (MMLU $27.4 \rightarrow 3.6\%$, about 24 pp; GPQA $25.0 \rightarrow 3.2\%$, about 22 pp). The collapse therefore does not require extreme compression, and its magnitude does not scale with how much compression occurred, which is evidence against reading the effect as a simple monotone function of trace length. The faithfulness collapse is not weak-to-strong-specific and not GPQA-specific.

The collapse is not an artifact of testing math-trained models on science. Restricting the MMLU replication to the `college_mathematics` subject (an in-domain evaluation, matched to the MATH-L3–L5 training data), the pattern is if anything more dramatic than on the science subjects. Baseline acknowledges 36.7% (11/30 matched (qid, cue) pairs), W2SR 2.5% (1/40), and Self-A 0.0% (0/40). Paired McNemar exact $p = 0.006$, $\Delta = -0.333$, discordant $11/1$ (Self-A vs. baseline: $p = 0.001$, $\Delta = -0.367$, discordant $11/0$). For contrast the combined non-math STEM subjects (biology, chemistry, physics variants) give baseline 25.6% $(41/160)$, W2SR 3.9% $(6/155)$. The dissociation does not require a distribution shift from training to evaluation: the SFT collapses cue acknowledgment even when we evaluate on math questions.

What does the CoT itself derive? The dissociation results so far tell us that the model's answer is silently cue-influenced and the judge sees no acknowledgment, but they do not tell us what the reasoning trace itself would predict if read alone. On the 255 W2SR samples with $\mathrm{influenced} = 1$, ack$= 0$, and cue_target $\neq$ correct, we strip the trailing `ANSWER: X` line and send the remaining CoT (along with the four answer choices, but no cue text) to a fresh judge (`claude-sonnet-4.5` at temperature 0) with the question "which letter is this reasoning arriving at?" The results (Table 3): in $214/255$ (83.9%) of samples the CoT itself concludes the cue target, either by explicitly naming it (185) or by deriving a value that maps to it (29). Only $14/255$ (5.5%) show the strong "silent flip" pattern where the CoT reasons to the correct answer and the emitted answer flips to the cue. The pattern is consistent across text cues (85.3%, 82.4%, 86.7% for grader-hack, insider-info, Stanford-professor respectively; the two non-text cues have $n < 20$ and are noisy). One mechanical caveat: the stripper removed only a trailing `ANSWER: X` line, and in 30 of the 255 CoTs the emitted letter also survived in another terminal form (a closing `\boxed{}` or a final "the answer is X" sentence), so the judge could in principle read the letter off the emission rather than the reasoning. Nearly all of the 30 were judged walks-to-cue (at least one was judged a silent flip); excluding every one of them still gives $184/225 = 81.8\%$, so the headline does not rest on them (and a CoT that closes by naming its own conclusion is largely what the transparent "by naming the letter" category is already reporting).

As a control, we ran the same judge on the 36 baseline R1-7B samples that meet the same influenced=1, cue_target $\neq$ correct filter (baseline naturally acknowledges the cue $\sim$25% of the time so the strict ack=0 restriction leaves too few samples to compare). Baseline's CoT also concludes the cue target in $28/36$ (77.8%) of these cases, and $9/10$ (90%) on the subset where baseline additionally did not acknowledge (matched to the W2SR condition but small n). So the "CoT walks to the cue when the model is influenced" pattern is not W2SR-specific: it is the default behavior of any silently-influenced model. What W2SR distinctly loses is not the reasoning direction, which is already corrupted whenever the cue moves the answer, but the acknowledgment overlay: baseline warns the overseer 25% of the time by explicitly mentioning the cue; W2SR warns only 3% of the time. The corrupted reasoning underneath is comparable; what changes is whether the trace surfaces the warning.

For monitoring, the practical implication sharpens rather than weakens: a naive overseer reading only the CoT of a cue-influenced completion would concur with the model's wrong answer regardless of the model (baseline or W2SR), because the CoT itself is walking to that wrong answer. The

Table 3: What the CoT itself derives on the safety-relevant W2SR subset ($n = 255$, $\text{influenced} = 1$, ack= 0, cue_target $\neq$ correct). Judge is Claude Sonnet 4.5 at $T = 0$; the trailing `ANSWER: X` line is removed from the CoT and the four answer choices are shown (no cue text, no question stem).

what the CoT reasoning arrives at	n	fraction
cue target (CoT walks to cue)	214	83.9%
by naming the letter	185	72.5%
by deriving a value that maps to it	29	11.4%
correct answer (silent flip: reasoning correct, answer flipped)	14	5.5%
derivation unmappable to any letter	13	5.1%
some third letter (neither correct nor cue)	12	4.7%
judge failed	2	0.8%

W2SR-specific harm is not that the CoT direction changed but that the model's tendency to comment on the cue, which is the overseer's only independent signal, was stripped away.

Is the acknowledgment capability recoverable at inference time? The dissociation results so far are silent on whether the acknowledgment capability is destroyed by SFT or only suppressed by default context. To distinguish, we ran the W2SR R1-7B model on ~ 100 matched (qid, cue) samples under four conditions: no system prompt (matching the paper's main-eval setup), a "think for at least 2000 tokens and consider all information including any hints" long-CoT prompt, an "if you notice any hints, cues, external suggestions, or biases in the prompt, name them in your reasoning" explicit-acknowledge prompt, and a few-shot in-context-learning (ICL) prompt that shows the model a worked example of a CoT that explicitly names an authority-opinion hint before verifying the answer independently, then asks it to solve the target problem in that same style. Pooled across cues:

condition	n	ack	rate	vs none (paired)	p
none (no system prompt)	99	1	1.0%	–	–
long-CoT prompt	99	6	6.1%	5 recovery-only, 0 none-only	0.06
acknowledge prompt	99	5	5.1%	4 recovery-only, 0 none-only	0.13
ICL example prompt	100	9	9.0%	8 recovery-only, 0 none-only	0.008

The ICL prompt is the strongest of the three (Figure 2B), reaches statistical significance under paired McNemar (8/0 discordant, $p = 0.008$), and in no case did any of the three prompts reduce acknowledgment relative to none.

Broken down by cue, the recovery concentrates almost entirely on one cue type. On `stanford_professor_recommends` (an authority-opinion cue), the ICL prompt lifts ack from $1/20$ (5%) to $8/20$ (40%), close to the baseline R1-7B level of 47% on that cue; long-CoT reaches $6/20$ (30%) and acknowledge reaches $4/20$ (20%). On `grader_hack_validation`, the two non-text cues, and mostly on `insider_information`, none of the three prompts produce measurable recovery (0–1 acks in each cell of 18–21 samples). Of the nine pooled ICL recoveries, eight come from the Stanford-professor cue alone. As a brief diagnostic, in the first 400 characters of the ICL completions on `grader_hack_validation`, none of the 21 samples in either the none or the ICL condition contain grader-related tokens ("grader," "hack," "validation," "answer key"), suggesting the model is not merely failing to flag a token it produces but is not producing the concept at all in its opening reasoning.

The reading is that the acknowledgment capability is not uniformly destroyed by SFT: for a highly nameable, salient authority cue the model can be prompted (with a demonstration example) to acknowledge it, reaching close to the untrained baseline rate on that cue. But for less-nameable cues (grader-hack framing, insider-information framing) even the strongest inference-time intervention we tested recovers essentially nothing. A system-prompt intervention is a partial mitigation restricted to prompt-elicitable cue types; it is not a substitute for training-side preservation of monitorability.

6 Discussion

Math-CoT SFT does not preserve monitorability, and where it is measurable, it hurts. The two substrates we look at are consistent in opposite directions relative to what an observer would conclude

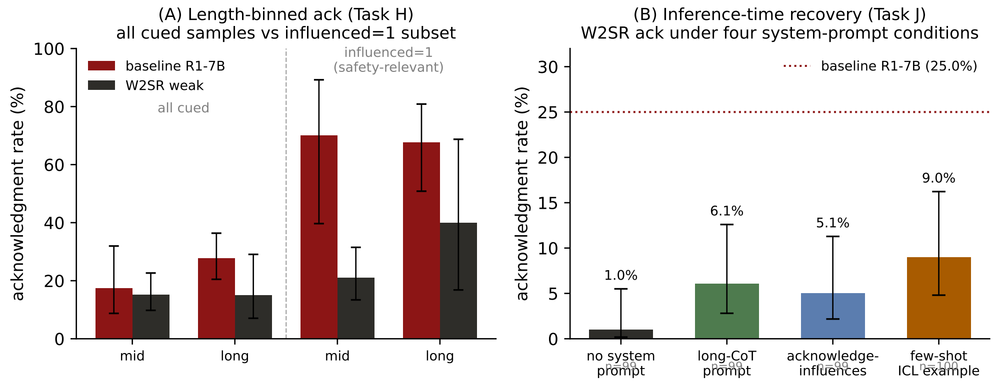

Figure 2: Left (A): length-binned acknowledgment (Task H), baseline R1-7B vs W2SR weak. On all cued samples (left half) the gap at matched mid-length is essentially closed; restricted to the safety-relevant $\text{influenced} = 1$ subset (right half) the gap reopens substantially. Error bars: Wilson 95% CI. Right (B): inference-time recovery (Task J). Ack rate on W2SR R1-7B under four system-prompt conditions at ~ 100 matched (qid, cue) pairs per condition. The few-shot ICL example prompt is the strongest and reaches statistical significance ($p = 0.008$, paired McNemar), but the pooled recovery still sits well below the baseline R1-7B rate of 25%; the recovery concentrates on the Stanford-professor cue (see text).

Table 4: Per-text-cue acknowledgment, baseline R1-7B vs. W2SR (paired on question id). Ack drops on every text cue, and each cue's discordant cells are one-directional (W2SR-only ack ≤ 1). The two non-text cues (`visual_squares`, `xml_metadata`) sit at 0% on both conditions and are omitted. Per-cue p-values are exploratory and uncorrected for multiplicity.

cue	baseline R1-7B	W2SR (1.5B)	paired Δ (n, disc. base/W2SR, p)
stanford_professor	$15/32 = 46.9\%$	$3/38 = 7.9\%$	-0.40 (30, $12/0$, 5×10^{-4})
grader_hack	$7/32 = 21.9\%$	$1/38 = 2.6\%$	-0.20 (30, $7/1$, 0.07)
insider_info	$18/32 = 56.2\%$	$2/38 = 5.3\%$	-0.50 (30, $15/0$, 6×10^{-5})

from Meek's composite measure. On the floored instruct student, Meek's composite "monitorability" (the mean of verbosity and faithfulness scores defined in [2]) rises after distillation almost entirely via verbosity. The model looks more monitorable to that measure without being more monitorable to the cue test. On the reasoning student, the cue's behavioral pull on the answer is not reduced while verbal acknowledgment collapses. Compression alone predicts most of that raw collapse; what a compression-only account does not predict is the exploratory matched-length residual on the $\text{influenced} = 1$ subset, where the model keeps being moved by the cue but stops reporting it at matched length. The safety-relevant component gets worse, not better.

The warning is sharp. An accuracy-only certification of any CoT-SFT student trained on MATH-style traces, including same-strength self-distillation rather than just weak-to-strong as in W2SR, can pass a model whose chain-of-thought has become markedly less revealing, and a monitorability check that rewards elaborate-looking CoT misses the same failure. The narrowness of the W2SR accuracy claim itself (in-family teacher and a base student, Section 4) sharpens this further. The certified guarantee is narrow on both axes.

7 Limitations

The supervision was answer-only. As described in Section 3, the student's chat template removed the reasoning span from every SFT example, so the R1-substrate arms were trained on the final-answer segments of math CoT traces rather than on the reasoning. This is the most important caveat on the paper and it bounds three claims. First, the CoT compression is explained by the supervision format and should not be read as an emergent property of CoT distillation. Second, the teacher-strength invariance is weaker evidence than it looks: with the teacher's reasoning excluded from the loss, weak, strong, and self teachers differ only in their final answers, so their near-identical effect is close to what one would predict. Third, and most consequentially, we cannot say from these experiments whether SFT that genuinely preserves the reasoning span would produce the same dissociation; that is the natural next experiment and we have not run it. What the results do establish is unaffected by the format question: these trained models retain the cue's behavioral pull while their traces stop reporting it, and they do so at matched length on the safety-relevant subset. The certification argument also survives intact, since answer-only SFT on math traces is itself a common and cheap post-training recipe, and it is exactly the kind of change an accuracy-only gate would wave through.

The instruct arm only motivates the substrate choice. The faithfulness claim rests on the reasoning student. The matched-length residual depends on the cut we take. Across all cued samples, the gap at matched mid-length is closed (Task H bin 3, 17.5% vs 15.1%, Fisher $p = 0.80$) and at long lengths is marginal (bin 4: $+12.7$ pp, OR ≈ 2.2, Fisher one-sided $p = 0.08$). On the safety-relevant subset, the matched-mid-length gap is 70.0% vs 21.1%, OR ≈ 8.75, $p = 0.003$, and at long lengths it is 67.6% vs 40.0% (underpowered, $n = 34$ vs 10). Compression accounts for most of the raw drop in absolute ack rates; cue-suppression accounts for the matched-length residual on the safety-relevant subset. We do not claim compression is the sole mechanism.

The degradation is shown within the Qwen R1-distill lineage, under weak, strong, and same-strength (self) teachers. A same-recipe cross-family test on R1-Distill-Llama-8B (self-distillation, matched LoRA config, 8k generation budget, same MATH L3 to L5 problem pool) fails the capability gate cleanly: GPQA accuracy goes from 50.0% to 22.5% (-27.5 pp at 0% empty completions on a verified-warm endpoint), at about $10.8\times$ CoT compression (15,420 to 1,424 chars median). This is substrate sensitivity, not infrastructure noise: the recipe that holds Qwen-7B capability while compressing about $14\times$ over-compresses Llama-8B and craters accuracy, putting the cross-family Self-B arm in the over-compression confound regime, so we do not run its monitorability eval. Cross-family generalization of the dissociation therefore remains open: at matched recipe, Llama fails the gate before the dissociation question is measurable.

The MMLU replication covers five STEM subjects; broader coverage, including humanities and social sciences, is untested, and the per-subject cells are small (30–40 cued samples each) even though the pooled comparison is well powered. Other limits: LoRA rather than full SFT; MATH-L3 to L5 SFT data only, with other domains untested; and judge robustness checked against two alternative families (`gemini-2.5-pro`, `kimi-k2-0905`) rather than a full panel. Absolute acknowledgment rates should be read as judge-dependent in any case [15]; what replicates across our three judges is the direction and the one-directional discordant pattern, not the level.

The $\text{influenced} = 1$ cut that carries the matched-length residual conditions on a post-treatment variable (whether the cue moved the answer), and it is one cell of a two-stratum by five-bin grid reported without multiplicity correction, with $n = 10$ on the baseline side. We report it because it is the subset monitorability is about, but it is the most fragile number in the paper and should be treated as a well-motivated exploratory finding rather than a confirmatory test.

Finally, the analysis plan was preregistered before any trained student existed (`PREREGISTRATION.md`). Several deviations should be noted. The preregistered primary estimand was W2SR $-$ strong-teacher control, which is null here ($p = 0.09$); our headline is the preregistered secondary estimand, W2SR $-$ baseline. Under the preregistered decision rules this is the H0 case, "no weak-supervision-specific effect, but an SFT-on-CoT effect that is itself reportable," which is what we report. The prereg also called for judge–human agreement on a hand-labeled set before trusting the acknowledgment labels; we substituted cross-judge agreement across three model families and a within-judge rubric swap, which is weaker on the human-validity axis. Smaller deviations: the preregistered combined monitorability score is not reported for the reasoning-student arms (we report its components separately), the per-cue comparisons in Table 4 are exploratory and uncorrected, and the optional mixed-effects robustness model was not run.

8 Conclusion

W2SR's accuracy gain is real but conditional, requiring an in-family teacher and a base student. On a reasoning student, its faithfulness collapse (25% to 3% at preserved accuracy) is not specific to weak-to-strong distillation: a same-strength self-distillation control reproduces the collapse at indistinguishable magnitude, and the cue's behavioral pull on the answer is not reduced while verbalization collapses. The dissociation survives a within-judge rubric swap (under which it tightens to acknowledgment $9.4\% \rightarrow 0.0\%$), two out-of-family judges, and a cross-substrate replication on MMLU. Most of the raw drop follows from the compression that answer-only supervision produces. What compression does not explain is the exploratory residual on the safety-relevant subset: restricted to samples where the cue actually moved the answer, acknowledgment is suppressed at matched length ($\mathrm{OR} \approx 8.75$, $p = 0.003$; §7), and the MMLU replication reproduces the full collapse at $2.8\times$ compression rather than GPQA's $14\times$, so the effect is not a simple function of how much shorter the traces got. A same-recipe cross-family attempt on R1-Distill-Llama-8B fails the capability gate (-27.5pp accuracy at $10.8\times$ compression), so the dissociation's cross-family scope remains open. Certifying any CoT-SFT student on accuracy alone, or on how elaborate its CoT looks, is insufficient and potentially misleading. The cue-acknowledgment component of monitorability must be measured directly.

Acknowledgments

Thanks to Modal for the GPU credits that supported trace generation, training, and serving. The Anthropic Claude, Google Gemini, and Moonshot Kimi judges (the latter two via OpenRouter) contributed to the cross-model and cross-rubric robustness analyses. All errors are my own.

References

[1] Yuan et al. Incentivizing Strong Reasoning from Weak Supervision (W2SR). arXiv:2505.20072, 2025.

[2] Meek et al. Measuring Chain-of-Thought Monitorability Through Faithfulness and Verbosity. arXiv:2510.27378, 2025.

[3] Turpin et al. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. arXiv:2305.04388, 2023.

[4] Chua & Evans. Are DeepSeek-R1 and Other Reasoning Models More Faithful? arXiv:2501.08156, 2025.

[5] Lobo, Agarwal, Lakkaraju. On the Impact of Fine-Tuning on Chain-of-Thought Reasoning. arXiv:2411.15382, 2024.

[6] Kim et al. Why Does Self-Distillation (Sometimes) Degrade the Reasoning Capability of LLMs? arXiv:2603.24472, 2026.

[7] Zaman, K. & Srivastava, S. Is Chain-of-Thought Really Not Explainability? Chain-of-Thought Can Be Faithful without Hint Verbalization. arXiv:2512.23032, 2025.

[8] Young, R. J. Lie to Me: How Faithful Is Chain-of-Thought Reasoning in Reasoning Models? arXiv:2603.22582, 2026.

[9] Korbak, T., Bengio, Y., Dragan, A., et al. Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety. arXiv:2507.11473, 2025.

[10] Yang, Y., Ma, Y., Liu, P. Weak-to-Strong Reasoning. Findings of EMNLP, 2024. arXiv:2407.13647.

[11] Jiang, Z., Peng, X., Teng, F., Fu, Z., Kim, Y., Mi, J., Li, Z., Wu, H. Better Accuracies, Worse Reasoning: A Step-Level Audit of Medical Chain-of-Thought Distillation. arXiv:2605.28301, 2026.

[12] Xiong, Z., Chen, S., Lakkaraju, H. Monitorability as a Free Gift: How RLVR Spontaneously Aligns Reasoning. arXiv:2602.03978, 2026.

[13] Yee, E., Li, A., Tang, C., Jung, Y. H., Paturi, R., Bergen, L. Dissociation of Faithful and Unfaithful Reasoning in LLMs. arXiv:2405.15092, 2024.

[14] Little, B. Length Penalties Make Chain-of-Thought Less Monitorable. arXiv:2607.09786, 2026.

[15] Young, R. J. Measuring Faithfulness Depends on How You Measure: Classifier Sensitivity in LLM Chain-of-Thought Evaluation. arXiv:2603.20172, 2026.

[16] Scalena, D., Candussio, S., Bortolussi, L., Fersini, E., Nissim, M., Sarti, G. Beyond the Commitment Boundary: Probing Epiphenomenal Chain-of-Thought in Large Reasoning Models. arXiv:2606.13603, 2026.

[17] Za, J., Bainiaksina, J., Ostrovsky, N., Chopra, T., Krakovna, V. Persuasion Attacks Can Decrease Effectiveness of CoT Monitoring. arXiv:2607.08066, 2026.

[18] Young, R. J. Why Models Know But Don't Say: Chain-of-Thought Faithfulness Divergence Between Thinking Tokens and Answers in Open-Weight Reasoning Models. arXiv:2603.26410, 2026.